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Course Code/Course Name: DSA06 /Data Handling and Visualization

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LIST OF EXPERIMENTS

Set 1: Monthly Sales Data

Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

Monthly Sales Data

```
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
```

```
sales <- c(120, 150, 180, 170, 210, 250,  
          240, 270, 300, 320, 350, 400)
```

Create Line Chart

```
plot(sales,  
     type = "o",  
     xaxt = "n",  
     xlab = "Month",  
     ylab = "Sales",  
     main = "Monthly Sales",
```

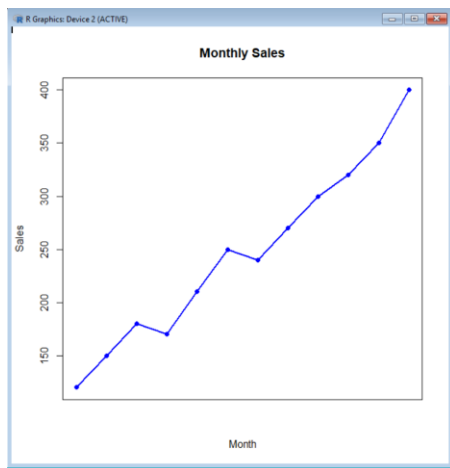
```
col = "blue",
```

```
pch = 16,
```

```
lwd = 2)
```

```
# Add Month Labels
```

```
axis(1, at = 1:12, labels = month)
```



2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

```
# Product Sales Data
```

```
products <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
sales <- c(500, 700, 450, 650, 800)
```

```
# Create Bar Chart
```

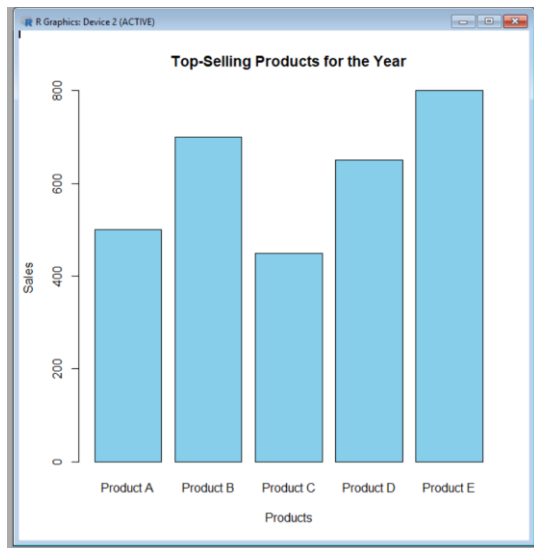
```
barplot(sales,
```

```
names.arg = products,
```

```
col = "skyblue",
```

```
main = "Top-Selling Products for the Year",
```

```
xlab = "Products",  
ylab = "Sales",  
border = "black")
```



3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

Explain the insights drawn from the scatter plot.

Advertising Budget and Monthly Sales Data

```
advertising <- c(10, 15, 20, 18, 25, 30, 28, 35, 40, 42, 45, 50)
```

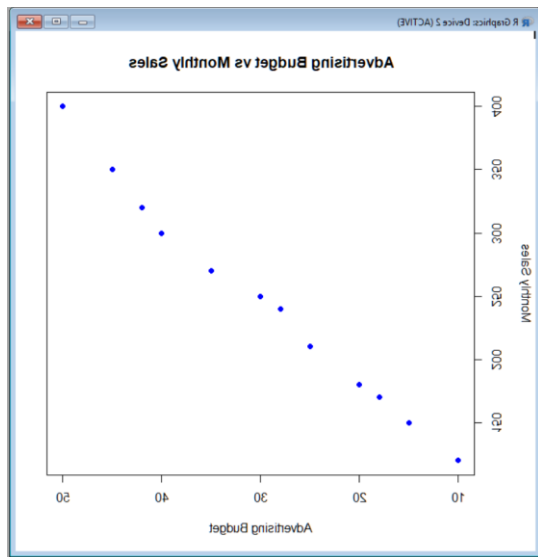
```
sales <- c(120, 150, 180, 170, 210, 250, 240, 270, 300, 320, 350, 400)
```

Create Scatter Plot

```
plot(advertising, sales,  
     main = "Advertising Budget vs Monthly Sales",  
     xlab = "Advertising Budget",  
     ylab = "Monthly Sales",  
     pch = 16,  
     col = "blue")
```

```
# Add Trend Line
```

```
abline(lm(sales ~ advertising), col = "red", lwd = 2)
```



4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore

sales data interactively.

```
# Install packages (Run only once)
```

```
# install.packages("flexdashboard")
```

```
# install.packages("plotly")
```

```
library(flexdashboard)
```

```
library(plotly)
```

```
# Monthly Sales Data
```

```
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
```

```

      "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

sales <- c(120,150,180,170,210,250,240,270,300,320,350,400)

# Product Sales Data

products <- c("Product A", "Product B", "Product C", "Product D", "Product E")

product_sales <- c(500,700,450,650,800)

# Line Chart

line_chart <- plot_ly(
  x = month,
  y = sales,
  type = "scatter",
  mode = "lines+markers"
) %>%
layout(
  title = "Monthly Sales",
  xaxis = list(title = "Month"),
  yaxis = list(title = "Sales")
)

# Bar Chart

bar_chart <- plot_ly(
  x = products,
  y = product_sales,
  type = "bar"
) %>%
layout(
  title = "Top-Selling Products",

```

```

xaxis = list(title = "Products"),
yaxis = list(title = "Sales")
)

```

```

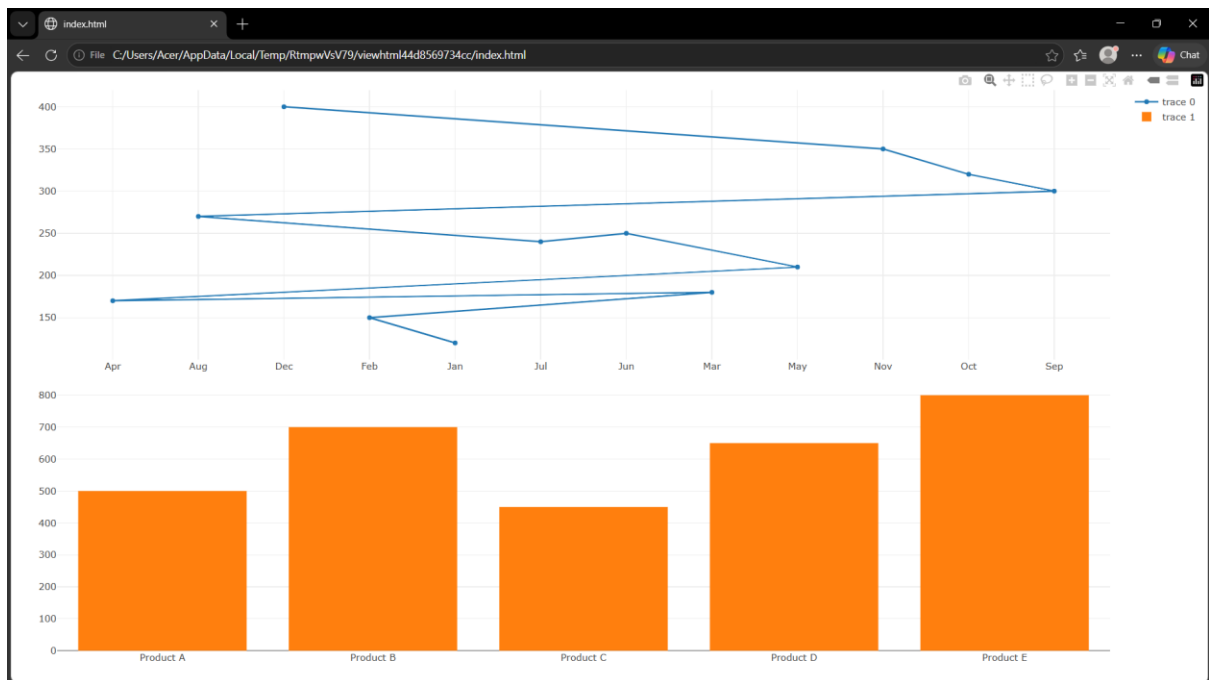
# Display Dashboard

```

```

subplot(line_chart, bar_chart, nrows = 2, shareX = FALSE)

```



Set 2:

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```

# Customer Age Data

```

```

age <- c(25, 30, 35, 28, 40)

```

```

# Create Histogram

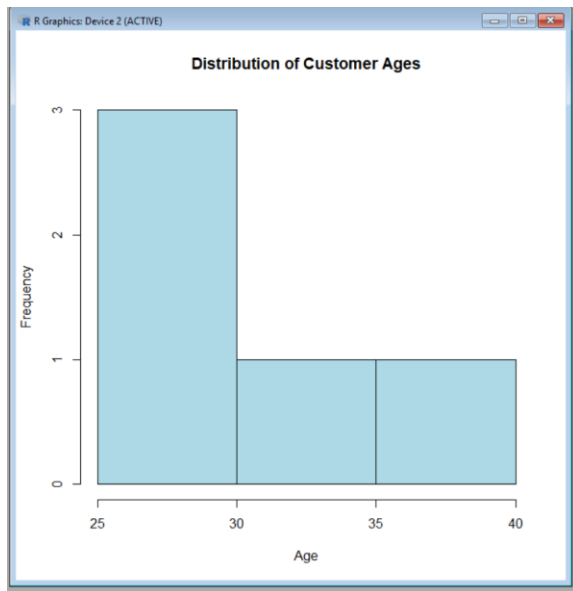
```

```

hist(age,
      main = "Distribution of Customer Ages",

```

```
xlab = "Age",  
ylab = "Frequency",  
col = "lightblue",  
border = "black",  
breaks = 5)
```



2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

```
# Customer Satisfaction Scores
```

```
satisfaction <- c(4, 5, 3, 4, 5)
```

```
# Count the frequency of each score
```

```
score_count <- table(satisfaction)
```

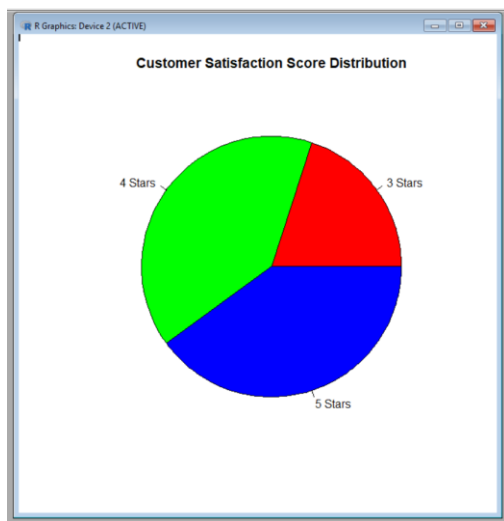
```
# Create Pie Chart
```

```
pie(score_count,
```

```
  labels = paste(names(score_count), "Stars"),
```

```
  main = "Customer Satisfaction Score Distribution",
```

```
  col = rainbow(length(score_count)))
```



3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
# Customer Data
```

```
age <- c(25, 30, 35, 28, 40)
```

```
satisfaction <- c(4, 5, 3, 4, 5)
```

```
# Create Age Groups
```

```
age_group <- c("20-29", "30-39", "30-39", "20-29", "40-49")
```

```
# Create Frequency Table
```

```
data <- table(age_group, satisfaction)
```

```
# Stacked Bar Chart
```

```
barplot(data,
```

```
  main = "Customer Satisfaction by Age Group",
```

```
  xlab = "Age Group",
```

```
  ylab = "Number of Customers",
```

```
  col = c("lightblue", "lightgreen", "pink"),
```



```
legend = rownames(data))
```



4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
# Install packages (Run only once)
```

```
install.packages("wordcloud")
```

```
install.packages("tm")
```

```
# Load libraries
```

```
library(wordcloud)
```

```
library(tm)
```

```
# Sample Customer Feedback
```

```
feedback <- c(
```

```
  "Excellent service",
```

```
  "Good quality product",
```

```
  "Fast delivery",
```

```
  "Excellent support",
```

```
  "Good experience"
```

```
)
```

```
# Create Corpus
```

```
corpus <- Corpus(VectorSource(feedback))
```

```
# Convert to plain text
```

```
corpus <- tm_map(corpus, content_transformer(tolower))
```

```
corpus <- tm_map(corpus, removePunctuation)
```

```
corpus <- tm_map(corpus, removeNumbers)
```

```
corpus <- tm_map(corpus, removeWords, stopwords("english"))
```

```
# Create Term Document Matrix
```

```
tdm <- TermDocumentMatrix(corpus)
```

```
m <- as.matrix(tdm)
```

```
words <- rownames(m)
```

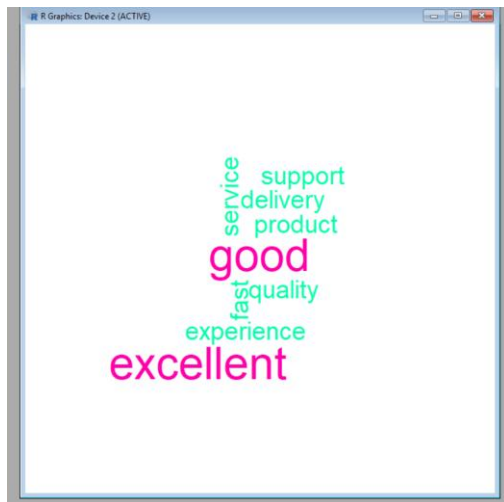
```
freq <- rowSums(m)
```

```
# Generate Word Cloud
```

```
wordcloud(words, freq,
```

```
  min.freq = 1,
```

```
  colors = rainbow(length(words)))
```



Set 3: Employee Performance Evaluation

Dataset: Employee Performance

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

Employee Data

```
employee <- c(1, 2, 3, 4, 5)
```

```
performance <- c(85, 92, 78, 90, 76)
```

Create Line Chart

```
plot(employee, performance,
```

```
  type = "o",
```

```
  col = "blue",
```

```
  pch = 16,
```

```
  lwd = 2,
```

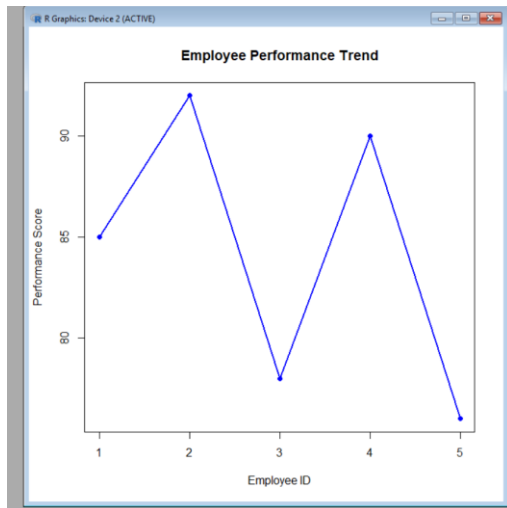
```
  main = "Employee Performance Trend",
```

```
  xlab = "Employee ID",
```

```
  ylab = "Performance Score")
```

```
# Add Legend
```

```
legend("topright",  
      legend = "Performance Score",  
      col = "blue",  
      lty = 1,  
      pch = 16)
```



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Employee Department Data
```

```
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
```

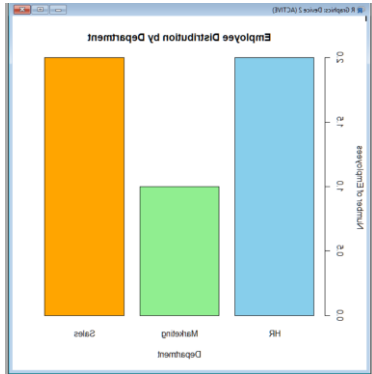
```
# Count Employees in Each Department
```

```
dept_count <- table(department)
```

```
# Create Bar Chart
```

```
barplot(dept_count,  
      main = "Employee Distribution by Department",  
      xlab = "Department",
```

```
ylab = "Number of Employees",
col = c("skyblue", "lightgreen", "orange"),
border = "black")
```



3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

```
# Employee Data
```

```
years <- c(5, 3, 7, 4, 2)
```

```
performance <- c(85, 92, 78, 90, 76)
```

```
# Create Scatter Plot
```

```
plot(years, performance,
```

```
  main = "Years of Service vs Performance Score",
```

```
  xlab = "Years of Service",
```

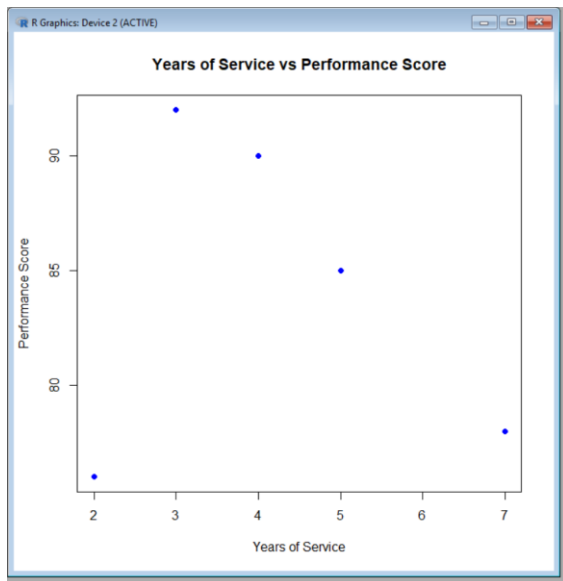
```
  ylab = "Performance Score",
```

```
  pch = 16,
```

```
  col = "blue")
```

```
# Add Trend Line
```

```
abline(lm(performance ~ years), col = "red", lwd = 2)
```



4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Steps to Create a Tableau Dashboard

- 1. Open Tableau Desktop.**
- 2. Import the Employee Performance dataset (Excel/CSV).**
- 3. Create the Line Chart:**
 - Drag **Employee ID** to **Columns**.
 - Drag **Performance Score** to **Rows**.
 - Change the **Marks** type to **Line**.
 - Add **Department** to **Color** to display a legend.
 - Add the title "**Employee Performance Trend**".
- 4. Create the Bar Chart:**
 - Drag **Department** to **Columns**.
 - Drag **Employee ID** to **Rows**.
 - Change the aggregation to **COUNT(Employee ID)**.
 - Change the **Marks** type to **Bar**.
 - Add the title "**Employee Distribution by Department**".

5. Create the Dashboard:

- Click **New Dashboard**.
- Drag the **Line Chart** and **Bar Chart** onto the dashboard.
- Arrange them side by side or one above the other.

6. Make it Interactive:

- Click **Dashboard → Actions → Add Action → Filter**.
- Set the **Line Chart** as the source and the **Bar Chart** as the target (or vice versa).
- Enable **Use as Filter** so selecting a department or employee updates the other chart.

Dashboard Features

-  Interactive **Line Chart** showing employee performance.
-  Interactive **Bar Chart** showing employee distribution by department.
-  Users can click, filter, and explore the employee performance data interactively.

Set 4: Product Inventory Management

Dataset: Product Inventory

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Product Inventory Data
```

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

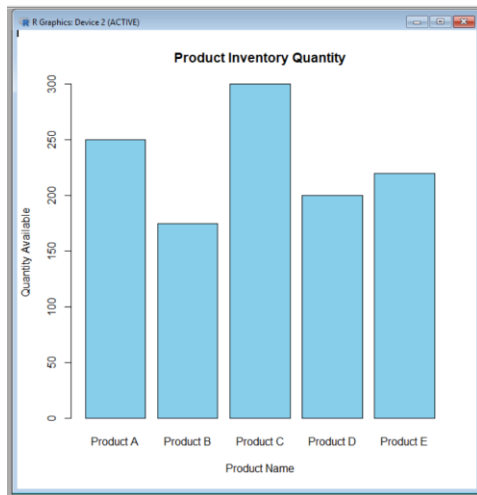
```
quantity <- c(250, 175, 300, 200, 220)
```

```
# Create Bar Chart
```

```
barplot(quantity,
```

```
  names.arg = product,
```

```
main = "Product Inventory Quantity",  
xlab = "Product Name",  
ylab = "Quantity Available",  
col = "skyblue",  
border = "black")
```



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

```
# Product Data
```

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
# Sample Categories
```

```
category <- c("Electronics", "Electronics", "Furniture", "Furniture", "Stationery")
```

```
# Quantity Available
```

```
quantity <- c(250, 175, 300, 200, 220)
```

```
# Create Data Table
```

```
inventory <- table(category, product)
```



```
# Multiply by quantity to show actual values
```

```
inventory[1,1] <- 250
```

```
inventory[1,2] <- 175
```

```
inventory[2,3] <- 300
```

```
inventory[2,4] <- 200
```

```
inventory[3,5] <- 220
```

```
# Stacked Bar Chart
```

```
barplot(inventory,
```

```
  main = "Product Quantity by Category",
```

```
  xlab = "Products",
```

```
  ylab = "Quantity Available",
```

```
  col = c("skyblue", "lightgreen", "orange"),
```

```
  legend = rownames(inventory))
```



3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
# Product Data
```

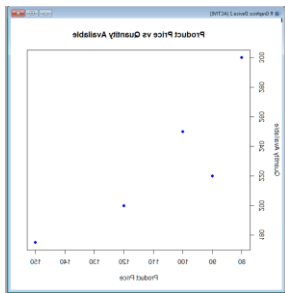
```
price <- c(100, 150, 80, 120, 90)
quantity <- c(250, 175, 300, 200, 220)
```

```
# Create Scatter Plot
```

```
plot(price, quantity,
      main = "Product Price vs Quantity Available",
      xlab = "Product Price",
      ylab = "Quantity Available",
      pch = 16,
      col = "blue")
```

```
# Add Trend Line
```

```
abline(lm(quantity ~ price), col = "red", lwd = 2)
```



4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Steps to Create the Dashboard

1. Open **Tableau Desktop**.
2. Import the **Product Inventory** dataset (Excel or CSV file).

Sheet 1: Bar Chart

- Drag **Product Name** to **Columns**.
- Drag **Quantity Available** to **Rows**.

- Select **Bar** from the **Marks** card.
- Add the title "**Product Inventory Quantity**".

Sheet 2: Stacked Bar Chart

- Drag **Product Name** to **Columns**.
- Drag **Quantity Available** to **Rows**.
- Drag **Product Category** (or the sample category field you created) to **Color**.
- Select **Bar** from the **Marks** card.
- Ensure **Analysis** → **Stack Marks** → **On** is enabled.
- Add the title "**Product Quantity by Category**".

Create the Dashboard

1. Click **Dashboard** → **New Dashboard**.
2. Drag both the **Bar Chart** and **Stacked Bar Chart** onto the dashboard.
3. Arrange them side by side or one above the other.

Make the Dashboard Interactive

- Click **Dashboard** → **Actions** → **Add Action** → **Filter**.
- Set one chart as the **Source** and the other as the **Target**.
- Enable **Run action on Select**.
- Users can now click on a product or category in one chart to filter and update the other chart interactively.

Dashboard Features

-  Bar Chart showing **Quantity Available** for each product.
-  Stacked Bar Chart showing **Quantity by Product Category**.
-  Interactive filtering and highlighting.
-  Easy comparison of product inventory across categories.

Set 5: Website Analytics

Dataset: Website Traffic

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

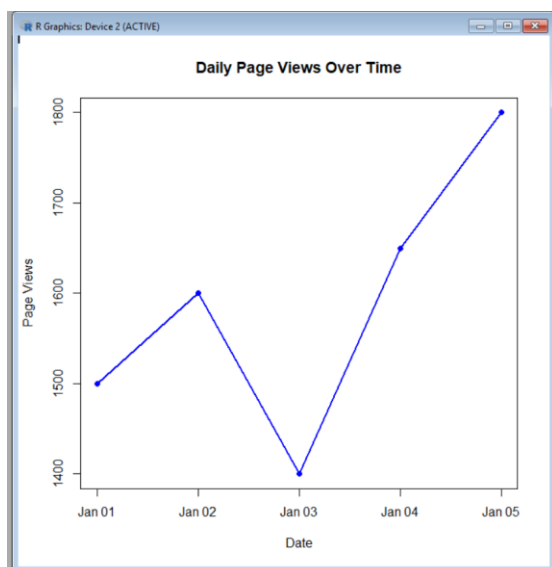
```
# Website Traffic Data
```

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",  
                 "2023-01-04", "2023-01-05"))
```

```
page_views <- c(1500, 1600, 1400, 1650, 1800)
```

```
# Create Line Chart
```

```
plot(date, page_views,  
     type = "o",  
     col = "blue",  
     pch = 16,  
     lwd = 2,  
     main = "Daily Page Views Over Time",  
     xlab = "Date",  
     ylab = "Page Views")
```



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Website Traffic Data
```

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03",  
         "2023-01-04", "2023-01-05")
```

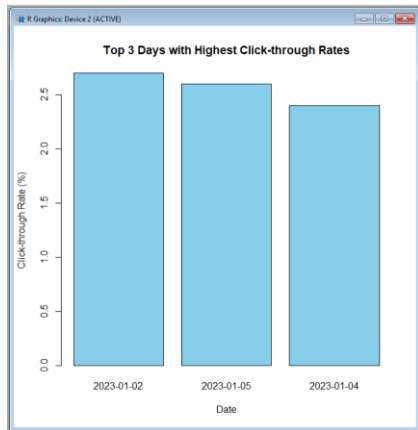
```
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)
```

```
# Select Top 3 Days with Highest CTR
```

```
top <- order(ctr, decreasing = TRUE)[1:3]
```

```
# Create Bar Chart
```

```
barplot(ctr[top],  
       names.arg = date[top],  
       main = "Top 3 Days with Highest Click-through Rates",  
       xlab = "Date",  
       ylab = "Click-through Rate (%)",  
       col = "skyblue",  
       border = "black")
```



3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

Install package (Run only once)

```
install.packages("ggplot2")
```

Load library

```
library(ggplot2)
```

Install package (Run only once)

```
install.packages("reshape2")
```

Load library

```
library(reshape2)
```

Website Interaction Data

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03",  
          "2023-01-04", "2023-01-05")
```

```
likes <- c(100, 120, 110, 130, 140)
```

```
shares <- c(40,45,35,50,55)
```

```
comments <- c(20,25,18,30,28)
```

```
# Create Data Frame
```

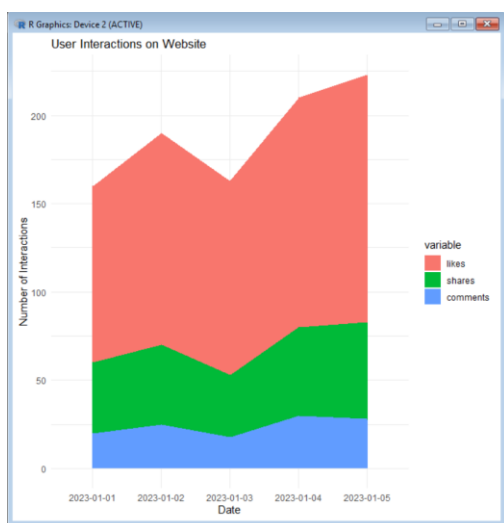
```
data <- data.frame(date, likes, shares, comments)
```

```
# Convert to Long Format
```

```
data_long <- melt(data, id.vars = "date")
```

```
# Create Stacked Area Chart
```

```
ggplot(data_long,  
       aes(x = date, y = value, fill = variable, group = variable)) +  
  geom_area() +  
  labs(title = "User Interactions on Website",  
       x = "Date",  
       y = "Number of Interactions") +  
  theme_minimal()
```



4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website

traffic data.

Steps to Create the Dashboard

1. Open **Tableau Desktop**.
2. Import the **Website Traffic** dataset (Excel/CSV).

Sheet 1: Line Chart

- Drag **Date** to **Columns**.
- Drag **Page Views** to **Rows**.
- Set the **Marks** type to **Line**.
- Add the title "**Daily Page Views Over Time**".

Sheet 2: Bar Chart

- Drag **Date** to **Columns**.
- Drag **Click-through Rate (CTR)** to **Rows**.
- Set the **Marks** type to **Bar**.
- Add the title "**Top Click-through Rates by Date**".

Create the Dashboard

1. Click **Dashboard** → **New Dashboard**.
2. Drag both the **Line Chart** and **Bar Chart** onto the dashboard.
3. Arrange them side by side or one above the other.

Make the Dashboard Interactive

- Click **Dashboard** → **Actions** → **Add Action** → **Filter**.
- Choose the **Line Chart** as the source and the **Bar Chart** as the target (or vice versa).
- Select **Run action on Select**.
- Now, clicking on a date in one chart filters and updates the other chart automatically.

Dashboard Features

-  Interactive **Line Chart** showing daily page views.

-  Interactive **Bar Chart** showing click-through rates.
-  Interactive filtering and highlighting.
-  Easy exploration and comparison of website traffic across different dates.

Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Product Sales Data

```
sales <- matrix(c(2000,2200,2400,
                 1500,1800,1600,
                 1200,1400,1100),
               nrow = 3, byrow = TRUE)
```

Product Names

```
rownames(sales) <- c("Product A", "Product B", "Product C")
```

Quarter Months

```
colnames(sales) <- c("January", "February", "March")
```

Create Grouped Bar Chart

```
barplot(sales,
       beside = TRUE,
       col = c("skyblue", "lightgreen", "orange"),
```

```
main = "First Quarter Product Sales",  
xlab = "Months",  
ylab = "Sales",  
legend.text = rownames(sales))
```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
# Install packages (Run only once)
```

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

```
# Load libraries
```

```
library(ggplot2)
```

```
library(reshape2)
```

```
# Sales Data
```

```
sales <- data.frame(  
  Month = c("January", "February", "March"),  
  Product_A = c(2000, 2200, 2400),  
  Product_B = c(1500, 1800, 1600),  
  Product_C = c(1200, 1400, 1600)
```

```
Product_C = c(1200, 1400, 1100)
)

# Convert data to long format
sales_long <- melt(sales, id.vars = "Month")
```

```
# Create Stacked Area Chart
ggplot(sales_long,
  aes(x = Month, y = value,
    fill = variable, group = variable)) +
  geom_area() +
  labs(title = "Overall Sales Trend for All Products",
    x = "Month",
    y = "Sales") +
  theme_minimal()
```



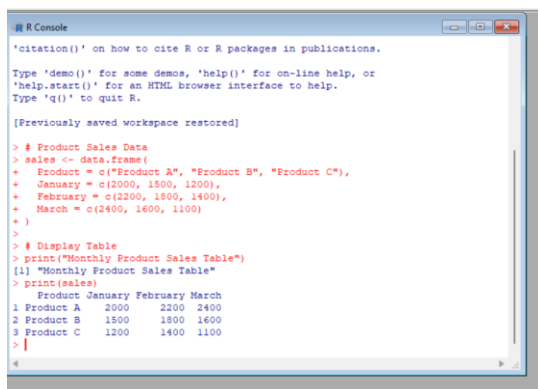
4. Build a table to show the monthly sales data for each product. Label the table elements.

```
# Product Sales Data
```

```
sales <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

```
# Display Table
```

```
print("Monthly Product Sales Table")  
print(sales)
```



```
R Console  
'citation()' on how to cite R or R packages in publications.  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.  
  
[Previously saved workspace restored]  
  
> # Product Sales Data  
> sales <- data.frame(  
+   Product = c("Product A", "Product B", "Product C"),  
+   January = c(2000, 1500, 1200),  
+   February = c(2200, 1800, 1400),  
+   March = c(2400, 1600, 1100)  
+ )  
>  
> # Display Table  
> print("Monthly Product Sales Table")  
[1] "Monthly Product Sales Table"  
> print(sales)  
  Product January February March  
1 Product A   2000    2200   2400  
2 Product B   1500    1800   1600  
3 Product C   1200    1400   1100  
> |  
4
```

4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Steps to Create the Dashboard

1. Open **Tableau Desktop**.
2. Import the **Monthly Product Sales** dataset (Excel/CSV).

Sheet 1: Grouped Bar Chart

- Drag **Month** to **Columns**.
- Drag **Sales** to **Rows**.

- Drag **Product Name** to **Color**.
- Select **Bar** from the **Marks** card.
- Add the title "**First Quarter Product Sales**".

Sheet 2: Stacked Area Chart

- Drag **Month** to **Columns**.
- Drag **Sales** to **Rows**.
- Drag **Product Name** to **Color**.
- Change the **Marks** type to **Area**.
- Ensure **Analysis** → **Stack Marks** → **On** is enabled.
- Add the title "**Overall Sales Trend**".

Sheet 3: Sales Table

- Create a new worksheet.
- Drag **Product Name** to **Rows**.
- Drag **January Sales**, **February Sales**, and **March Sales** to **Text**.
- Add the title "**Monthly Product Sales Table**".

Create the Dashboard

1. Click **Dashboard** → **New Dashboard**.
2. Drag the **Grouped Bar Chart**, **Stacked Area Chart**, and **Sales Table** onto the dashboard.
3. Arrange them neatly for easy viewing.

Make the Dashboard Interactive

- Click **Dashboard** → **Actions** → **Add Action** → **Filter**.
- Set any chart as the **Source** and the other charts/table as the **Target**.
- Choose **Run action on Select**.
- Clicking on a product or month in one visualization updates the others automatically.

Dashboard Features

-  **Grouped Bar Chart** to compare product sales across months.
-  **Stacked Area Chart** to show the overall sales trend.

-  **Sales Table** displaying monthly sales values.
-  Interactive filtering, highlighting, and comparison of product sales data.